

CLASS 03 / SEP 8 / U03

GEOMETRY FROM SCRATCH

P1 CONTINUES — WHAT A SOLID IS MADE OF, UNDERNEATH

NOTHING DUE

P1b + P1c ISSUED TODAY

POINT → LINE → EDGE → FACE → SOLID

Five steps, and each one is a list of the thing before it.

POINT	LINE	EDGE	FACE	MESH
three numbers	two positions	two vertices	ordered corners	many faces
[x, y, z]	[p1, p2]	[v1, v2]	[v1, v2, v3]	[f1, f2, f3...]

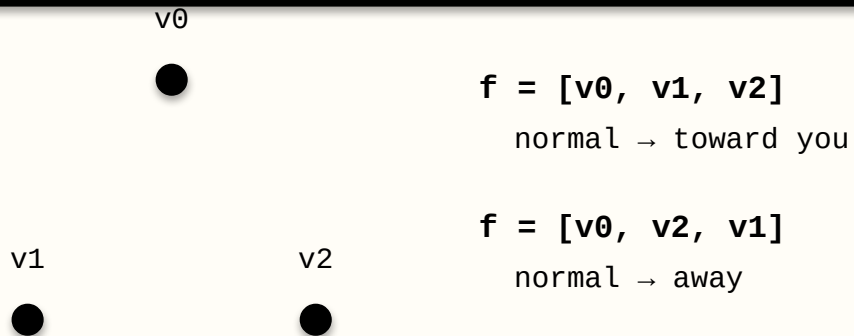
A POSITION IS NOT A DISPLACEMENT. THE SAME THREE NUMBERS MEAN DIFFERENT THINGS DEPENDING ON WHICH ONE YOU MEANT.

Nothing here is new geometry. It is the same collection, read at a different level: a face is a list of vertices, a mesh is a list of faces, and next week an array will be a list of copies.

A FACE IS AN ORDERED LIST OF CORNERS

Same three vertices, two different faces. The order is the difference.

WINDING ORDER



WHY IT MATTERS LATER

A flipped face tells a renderer the inside is the outside.

It tells a slicer there is nothing to fill.

Most failed prints this semester will start here, not at the printer.

CLOSED MEANS EVERY EDGE IS SHARED BY EXACTLY TWO FACES. CHECK IT BEFORE YOU BELIEVE IT.

THE PYRAMID

Five vertices, eight edges, five faces. Built by hand, driven by three numbers.

WHAT YOU ASSEMBLE

5

VERTICES

four base corners and an apex

8

EDGES

four around the base, four to the apex

5

FACES

four triangular sides and the base

EXPOSED INPUTS

Width

Depth

Height

Three numbers, and the whole solid rebuilds. That is the entire point of the exercise.

DO NOT FORGET THE BASE FACE. A PYRAMID WITH FOUR SIDES AND NO BOTTOM IS NOT A SOLID.

AND THEN IT GOES BACK INTO LAST WEEK'S FILE

The pyramid is not an exercise. It is the cutter for the study you already built.

01 BUILD IT

Five vertices, five faces, closed, with Width / Depth / Height exposed.

02 SUBTRACT IT

Cube minus pyramid. The same Boolean from Week 2, with your own solid.

03 DRIVE IT

Change the three inputs and show at least three settings of the result.

04 EXPLAIN IT

Face order, normals, closure — why it cuts cleanly instead of failing.

THIS IS WHY P1 IS ONE PROJECT AND NOT THREE ASSIGNMENTS.

Every project in this course does the same thing — the geometry you make in one milestone becomes the input to the next. You will never start from an unrelated object.

THE SAME THING, WRITTEN OUT

A mesh is two lists. That is the entire idea, and it fits on one slide.

```
verts = [[-w, -d, 0],          # 0
          [ w, -d, 0],          # 1
          [ w,  d, 0],          # 2
          [-w,  d, 0],          # 3
          [ 0,  0, h]]          # 4 apex

faces = [[0, 1, 4], [1, 2, 4],
          [2, 3, 4], [3, 0, 4],
          [0, 3, 2, 1]]         # the base – do not forget it

pyramid = mesh(verts, faces)
result  = difference(cube, pyramid)
```

WHAT IT MAPS ONTO

point	edge
vector	face
parameter	mesh
vertex	solid
line	normal

verts IS A LIST OF POSITIONS. faces IS A LIST OF INDICES INTO IT.

ISSUED TODAY

P1b — MESH FROM SCRATCH

DUE WEEK 5 — Sep 22 / 7%

Build the solid from its parts, then take it back to the massing study and let it do work there. One sheet covers the whole ladder — point, line, edge, face, mesh — and ends with an annotated closed pyramid.

SHOW

THE LADDER

Point, line, edge, face, and the multiple-face exercise.

BUILD

THE PYRAMID

Width, Depth and Height exposed. Four sides and a base.

REUSE

AS A CUTTER

Back into the P1a study, at three parameter settings.

REQUIREMENTS — P1b

Visual quality 35% · computational understanding 25% · technical execution 20% · experimentation 10% · requirements and identity 10%.

01

One sheet showing point, line, edge, face and the multiple-face mesh exercise, with an annotated closed pyramid.

02

Build four base corners and an apex with Width, Depth and Height exposed; include all four side faces and the base.

03

Reuse the pyramid as a Boolean cutter in the previous study and show at least three parameter settings. Explain face order, normals and closure.

ALSO ISSUED

P1c — FIRST PRINT

DUE WEEK 8 — Oct 13 / 7%

The same geometry, made real. A printable variant of the project you have been building for two weeks — presented as a physical object at the midterm.

LIMIT

60 MM, 200 G

Each axis within 60 mm. The allocation includes failures.

RULE

NO SUPPORTS

Orientation is your only lever. Change the geometry if you must.

DUE

AT THE MIDTERM

The queue has latency the two-week cycle does not.

REQUIREMENTS — P1c

Fabrication milestone — visual documentation 20% · computational understanding 20% · technical execution 15% · fabrication quality 30% · experimentation 5% · requirements 10%.

01

Develop a printable variant of the geometry project, within 60 mm on each axis and the 200 g allocation, with no supports.

02

Present the physical print at midterm with one orientation/process sheet and a photograph.

03

Submit the editable source, exported mesh and slicer project. Explain any geometry changes made for printability.

CHECKPOINT · WEEK 5 — Sep 22 — Review print-ready geometry, physical units, orientation and the sliced preview before queueing.

NOW — BUILD THE PYRAMID

Rest of the session: from a single point to a closed solid that cuts.

✓

One point placed from an XYZ vector, and you can say why it is a position

✓

Four base corners and an apex, with Width / Depth / Height exposed

✓

All five faces joined — four sides and the base

✓

Closure checked: every edge shared by exactly two faces

✓

Cube minus pyramid working, off the same three inputs

LEAVE WITH A SOLID THAT CUTS. THE SHEET IS THE EASY PART.

BEFORE NEXT CLASS

WATCH

U04 — Arrays and lists. Repeat Zone, Compare and Switch, and the nested grid. We start small and inspectable before anything scales up.

FINISH

P1a is due next class: the ordered diagram, the final axonometric and three variants.

BRING

Your pyramid, parametric and closed. Next week it becomes the component that gets repeated, so bring a file you can build on.

CHECK

Your FabLab orientation, before you need the machine rather than after.